Group Project – Examples

Group

Students choose a topic relevant to real-world investment management and vibe-code an end-to-end solution that leverages LLM as needed. The topic may be one covered explicitly in class or not. The proposals below span buy-side and sell-side use cases and are scoped so a group can get a working prototype running with public data.

1. Index Reconstitution Predictor from Methodology Documents. *Index providers publish their rules in prose (S&P U.S. Indices methodology, Russell reconstitution rules, or a thematic ETF's rulebook), and the index-inclusion effect around reconstitution is a well-documented trade. Students have Claude read the actual methodology PDF and generate executable Python screening logic (eligibility, size and liquidity thresholds, buffer rules, sector balance), verify the generated code against historical constituent changes, then run it ahead of the next reconstitution to predict additions and deletions and backtest a strategy around announcements. The evaluation covers both prediction accuracy and where the document's ambiguities (committee discretion, for instance) required human judgment the code could not capture.*

2. Fund Due Diligence and Style-Drift Detector. *Advisors and model-portfolio teams select third-party funds based on what the manager says they do; the holdings reveal what they actually do. Students build a due-diligence tool that uses Claude to extract a fund's stated strategy, benchmark, and constraints from its prospectus and factsheet, then pulls holdings from N-PORT filings on EDGAR (or ETF issuer files) to compute active share, factor exposures from the factor-model session, and a time series of style metrics. The engine flags closet indexing, style drift relative to the stated mandate, and concentration or liquidity concerns, and Claude writes a due-diligence memo that reconciles the narrative with the numbers. Evaluation covers how often stated and revealed strategy diverge across a fund universe and whether drift flags precede performance changes; the project connects factor modeling to a decision wealth managers make constantly.*

3. Earnings Call Guidance Extraction and Post-Earnings Drift Strategy. *Post-earnings announcement drift is one of the oldest documented anomalies, and the question is whether an LLM reading the call adds information beyond the headline earnings surprise. Students use Claude with structured outputs to extract from transcripts (free sources such as Motley Fool, or a transcript API) the guidance change versus prior guidance, management tone, and Q&A evasiveness, then build an event-driven long-short strategy on those scores with a proper event-study methodology: point-in-time transcript availability, cumulative abnormal returns over fixed windows, and controls for the standardized earnings surprise and size. The backtesting session's lessons on look-ahead bias and multiple comparisons apply directly, since groups will be tempted to test many extracted features. The deliverable is a signal, a backtest with controls, and an assessment of how much of the return comes from the LLM-derived features versus the earnings surprise itself.*

4. Central Bank Communication Index and Duration Positioning. *Students build a hawkish-dovish index from FOMC statements, minutes, press-conference transcripts, and governor speeches on the Federal Reserve site, using Claude to score each document on a defined rubric with evidence spans, and compare the score to simpler baselines such as dictionary methods and sentence-level embeddings. The index is tested against subsequent Treasury yield changes from FRED across the curve, and then used to drive a duration tilt in a Treasury or aggregate-bond portfolio with a backtest that accounts for the fact that the market moves on the statement within minutes, so the tradable edge, if any, lies in minutes, speeches, and the tone shift between consecutive statements.*

5. Investment Mandate Compliance Monitor. *Every institutional portfolio runs under a document — a fund prospectus, a sub-advisory agreement — that states constraints in prose: sector and issuer concentration caps, prohibited securities, minimum credit ratings, derivatives limits, ESG exclusions. Students take a real public prospectus and SAI from EDGAR (or write a realistic Investment Purpose Statement), use Claude to extract each constraint into a machine-checkable rule schema with a flag for rules that need human interpretation, and build a pre-trade/post-trade engine that checks a portfolio and a proposed trade list against the rulebook using holdings and reference data. Claude's second role is explaining each breach in plain language and proposing compliant remediation trades. This is one of the most widespread LLM deployments in asset-management operations.*

6. Reverse Stress Testing Agent. *Conventional stress testing starts from a scenario and computes the loss; reverse stress testing starts from a loss ("what would cost this portfolio 25%?") and searches for the most plausible scenarios that produce it, a regulatory expectation for banks and a useful exercise for any fund. Students build an agentic loop in which Claude proposes narratively coherent macro scenarios, maps each to a structured shock vector (rates, credit spreads, equity factors, FX, commodities) via a schema, and a factor-based risk engine computes the P&L; the results feed back so the agent refines toward scenarios that breach the threshold while staying plausible given historical covariance. The deliverable is a ranked set of scenarios with narratives, shock vectors, P&L decomposition, and an audit log of the agent's search. It inverts the scenario-analysis session and forces groups to think about how to keep an LLM from generating fantasy.*

7. Direct Indexing and Tax-Loss Harvesting Engine. *Direct indexing is the fastest-growing product in wealth management, and it is fundamentally an index-tracking problem with tax and client-customization constraints layered on top. Students build a separately-managed-account engine that tracks a benchmark with a sparse portfolio (regularized optimization), harvests losses at each rebalance, and selects replacement securities with matching factor and sector exposure. Claude's role is to translate the wash-sale and holding-period rules in IRS Publication 550 into rule logic the optimizer respects, to parse client restrictions stated in natural language ("no tobacco, keep my employer stock, I'm in the top bracket in New York") into constraints, and to generate the client-facing explanation of each harvest trade. Evaluation uses after-tax tracking difference and tax alpha against a plain ETF across different lot structures and market paths; a key verification point is which tax rules the LLM interpreted correctly and which needed hand-coding.*